

Artificial Intelligence – Lab

Project Document (Spring 2025)



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Dated

Income Prediction Using Logistic Regression on the UCI Adult Dataset

Abstract

This project aims to predict whether a person's income exceeds \$50K/year based on demographic and employment-related features using the UCI Adult Income dataset. The algorithm applied was **Logistic Regression**, a simple yet effective classification model. The process began with exploratory data analysis (EDA), followed by handling missing values, label encoding for categorical variables, and feature scaling using StandardScaler. The model achieved high accuracy, and the results indicate that features like education, occupation, and age have strong correlations with income. Data visualization techniques such as countplots, histograms, and correlation heatmaps provided deeper insights. The project concludes that Logistic Regression performs well for binary classification on this dataset.

Introduction

The problem addressed in this project is the **prediction of income class** (greater than \$50K or not) using individual demographic attributes. Predicting income class is valuable for marketing, finance, and social policy planning. The dataset used is the **UCI Adult Income dataset**, which contains features like age, workclass, education, marital status, occupation, relationship, race, sex, hours per week, and native country.

We chose **Logistic Regression** for this binary classification task because it's easy to interpret and computationally efficient. Logistic Regression works by applying a sigmoid function to model the probability of the binary outcome. It assumes a linear relationship between the independent variables and the log-odds of the dependent variable. While it may not handle non-linearity well, it's an excellent baseline model and performs strongly with clean and well-preprocessed data.

Methodology

1. **Data Cleaning:**
 - a. Replaced missing values represented by '?' with NaN and filled them using mode.
 - b. Checked for and ensured no remaining null values.
2. **Preprocessing:**
 - a. Label encoded categorical variables like workclass, education, occupation, etc.
 - b. Applied StandardScaler for feature scaling to standardize the numeric features.
3. **EDA (Exploratory Data Analysis):**
 - a. Countplot for income distribution.
 - b. Histogram of age by income class.
 - c. Bar chart comparing education level vs income.
 - d. Correlation heatmap for numerical features.
4. **Model Training:**
 - a. Used train_test_split (70% training, 30% testing).
 - b. Trained a Logistic Regression model using sklearn.linear_model.LogisticRegression.
 - c. Evaluated with accuracy and other performance metrics.
5. **Hyperparameters:**
 - a. Default hyperparameters were used as Logistic Regression does not require extensive tuning for baseline evaluation.

Results

The Logistic Regression model achieved the following performance metrics:

Model	Accuracy	Precision	Recall	F1 score	ROC auc
Logistic Regression	0.8453	0.7621	0.6923	0.7257	0.8834

Code

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JupyterLab Python [conda env:base]

```
[24]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

```
[26]: # View basic info
print(df.head())
print(df.shape)
print(df.info())
```

```

0    90  Private  77053    HS-grad         9    Widowed \
1    82  Private 132870    HS-grad         9    Widowed
2    66  Private 186061  Some-college      10    Widowed
3    54  Private 140359    7th-8th         4    Divorced
4    41  Private 264663  Some-college      10    Separated

   occupation relationship  race  sex  capital.gain \
0  Prof-specialty  Not-in-family  White  Female         0
1  Exec-managerial  Not-in-family  White  Female         0
2  Prof-specialty    Unmarried  Black  Female         0
3  Machine-op-inspct  Unmarried  White  Female         0
4  Prof-specialty    Own-child   White  Female         0

   capital.loss  hours.per.week  native.country  income
0         4356             40    United-States  <=50K
1         4356             18    United-States  <=50K
2         4356             40    United-States  <=50K
3         3900             40    United-States  <=50K
4         3900             40    United-States  <=50K
(32561, 15)
<class 'pandas.core.frame.DataFrame'>
```

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JupyterLab Python [conda env:base]

Loading the dataset

```
[29]: df = pd.read_csv("adult.csv")

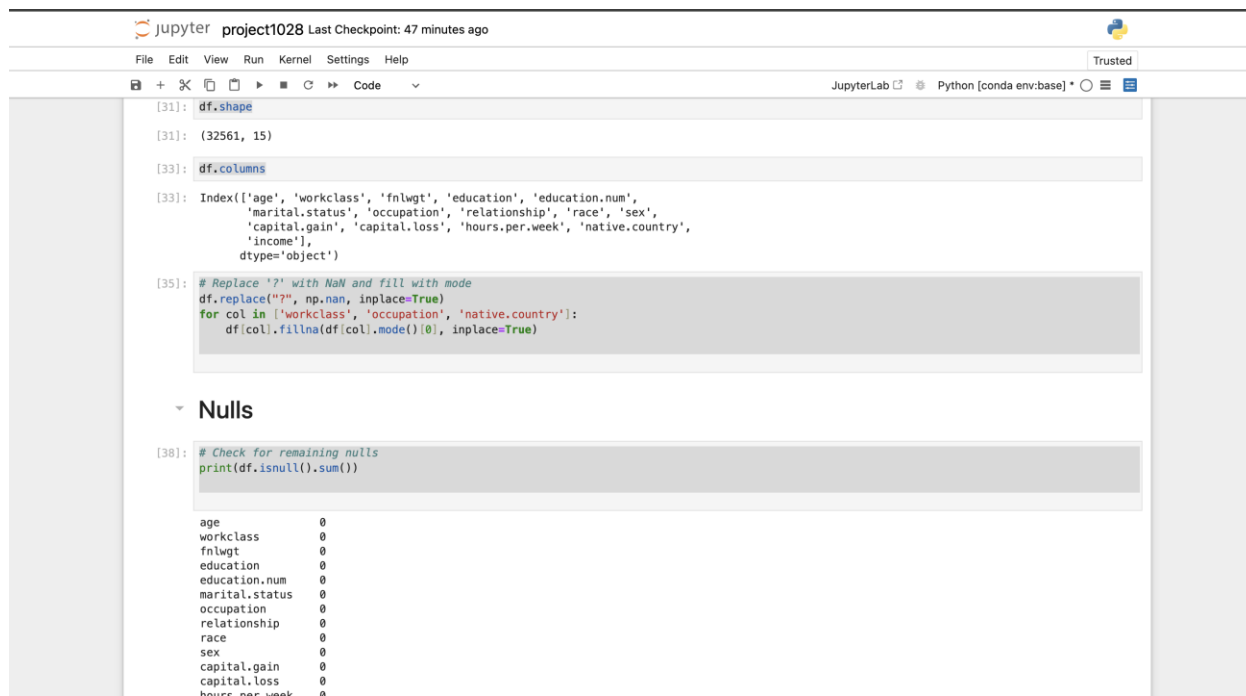
df.info()
df.head()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32561 entries, 0 to 32560
Data columns (total 15 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   age                 32561 non-null  int64
1   workclass           32561 non-null  object
2   fnlwgt              32561 non-null  int64
3   education           32561 non-null  object
4   education.num       32561 non-null  int64
5   marital.status      32561 non-null  object
6   occupation          32561 non-null  object
7   relationship        32561 non-null  object
8   race                32561 non-null  object
9   sex                 32561 non-null  object
10  capital.gain         32561 non-null  int64
11  capital.loss         32561 non-null  int64
12  hours.per.week       32561 non-null  int64
13  native.country       32561 non-null  object
14  income              32561 non-null  object
dtypes: int64(6), object(9)
memory usage: 3.7+ MB
```

```
[29]:
```

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relationship	race	sex	capital.gain	capital.loss	hours.per.week	native.c
0	90	?	77053	HS-grad	9	Widowed	?	Not-in-family	White	Female	0	4356	40	United-
1	82	Private	132870	HS-grad	9	Widowed	Exec-managerial	Not-in-family	White	Female	0	4356	18	United-
2	66	?	186061	Some-college	10	Widowed	?	Unmarried	Black	Female	0	4356	40	United-



The image shows a JupyterLab interface with a file explorer on the left, a top menu bar, and a central code editor. The code editor contains three code cells. The first cell runs `df.shape` and returns `(32561, 15)`. The second cell runs `df.columns` and returns a list of column names: `age, workclass, fnlwgt, education, education.num, marital.status, occupation, relationship, race, sex, capital.gain, capital.loss, hours.per.week, native.country, income`. The third cell runs a series of commands to replace '?' with NaN and fill NaN values with the mode for 'workclass', 'occupation', and 'native.country'. Below the code cells, there is a section titled 'Nulls' which shows the output of `df.isnull().sum()`, indicating that there are no null values in any of the columns.

```
[31]: df.shape
[31]: (32561, 15)
[33]: df.columns
[33]: Index(['age', 'workclass', 'fnlwgt', 'education', 'education.num',
        'marital.status', 'occupation', 'relationship', 'race', 'sex',
        'capital.gain', 'capital.loss', 'hours.per.week', 'native.country',
        'income'],
        dtype='object')
[35]: # Replace '?' with NaN and fill with mode
df.replace("?", np.nan, inplace=True)
for col in ['workclass', 'occupation', 'native.country']:
    df[col].fillna(df[col].mode()[0], inplace=True)

Nulls

[38]: # Check for remaining nulls
print(df.isnull().sum())

age          0
workclass    0
fnlwgt       0
education    0
education.num 0
marital.status 0
occupation   0
relationship 0
race         0
sex          0
capital.gain 0
capital.loss 0
hours.per.week 0
```

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JupyterLab Python [conda env:base]

```
y = df['income']
```

Train

```
[43]: # Split into train and test sets
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=0)
```

```
[45]: # Encode categorical features
from sklearn import preprocessing
categorical = ['workclass', 'education', 'marital.status', 'occupation', 'relationship', 'race', 'sex', 'native.country']

for feature in categorical:
    le = preprocessing.LabelEncoder()
    X_train[feature] = le.fit_transform(X_train[feature])
    X_test[feature] = le.transform(X_test[feature])
```

```
[47]: # Feature scaling
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train = pd.DataFrame(scaler.fit_transform(X_train), columns=X_train.columns)
X_test = pd.DataFrame(scaler.transform(X_test), columns=X_test.columns)
```

Algorithm

```
[50]: # Logistic Regression
```

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JupyterLab Python [conda env:base]

Algorithm

```
[50]: # Logistic Regression
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

logreg = LogisticRegression()
logreg.fit(X_train, y_train)
y_pred = logreg.predict(X_test)

print('Logistic Regression accuracy score with all features: {:.4f}'.format(accuracy_score(y_test, y_pred)))

Logistic Regression accuracy score with all features: 0.8218
```

```
[ ]:
```

Plot

1-Distribution of Income Classes

```
[56]: sns.countplot(x='income', data=df)
plt.title('Income Distribution')
plt.xlabel('Income Class')
plt.ylabel('Count')
plt.show()
```



